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数据结构与程序设计

(Python语言)

DATA STRUCTURE AND PROGRAM DESIGN IN PYTHON

安装Python编程环境 (Anaconda)



Python课程组



2024年09月



CONTENTS

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- 1 下载Anaconda安装文件
- 2 在Windows中安装并试用
- 3 在macOS中安装并试用



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01

下载安装文件



下载安装文件



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■ 三种下载方式

- ✿ 在QQ群下载老师上传的安装文件

- ✿ 在Anaconda官网下载

 - <https://repo.anaconda.com/archive/>

- ✿ 在清华大学镜像网站下载

 - <https://mirrors.tuna.tsinghua.edu.cn/anaconda/archive/>



下载安装文件



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选择安装文件

✱ 选择稳定的、使用人数较多的版本，不建议零基础的同学下载最新版本

Anaconda3-2023.07-2-MacOSX-arm64.pkg



使用苹果自研CPU (M1、M2、...) 的苹果电脑

Anaconda3-2023.07-2-MacOSX-arm64.sh

Anaconda3-2023.07-2-MacOSX-x86_64.pkg



使用英特尔CPU的苹果电脑 (一般是2021年前生产的)

Anaconda3-2023.07-2-MacOSX-x86_64.sh

Anaconda3-2023.07-2-Windows-x86_64.exe



操作系统是 Windows 的电脑



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02

在Windows中安装并试用



执行安装文件



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■ 在资源管理器中找到并执行安装文件

✦ Anaconda3-2023.07-2-Windows-x86_64.exe

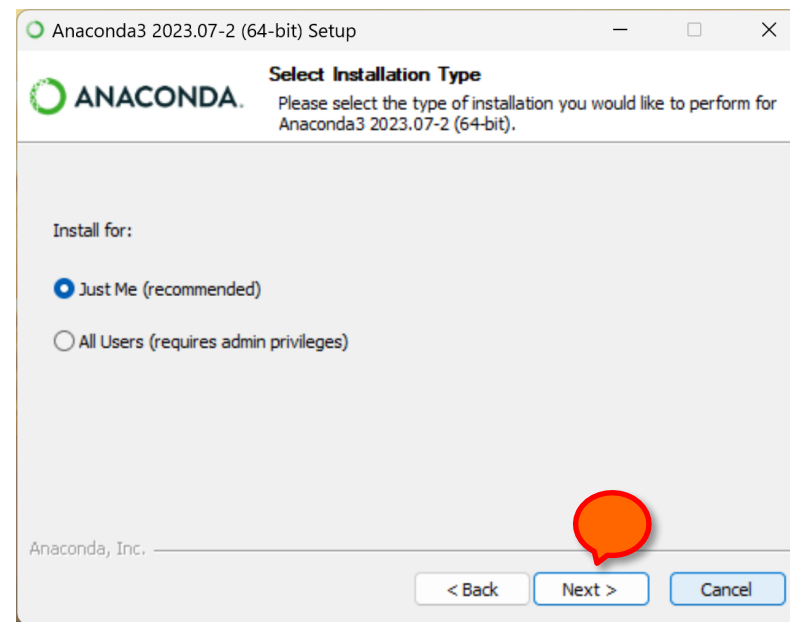
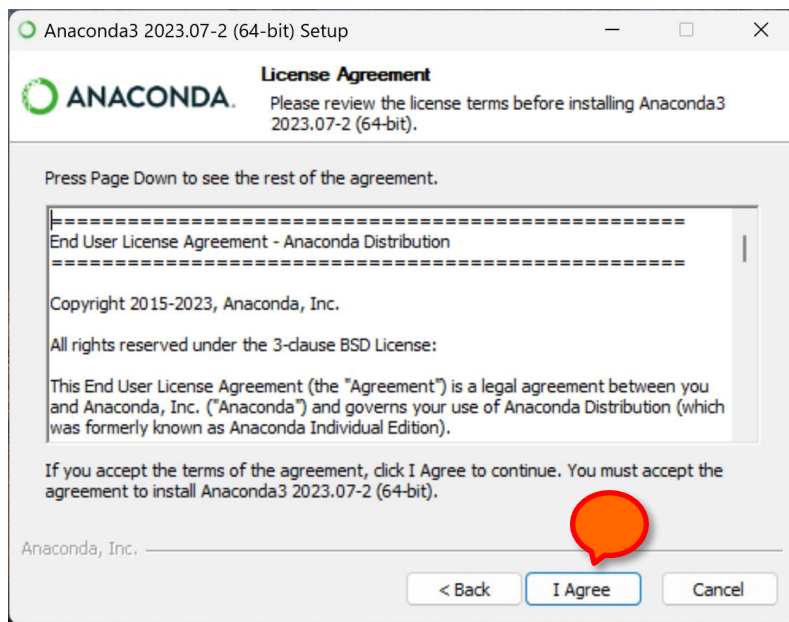
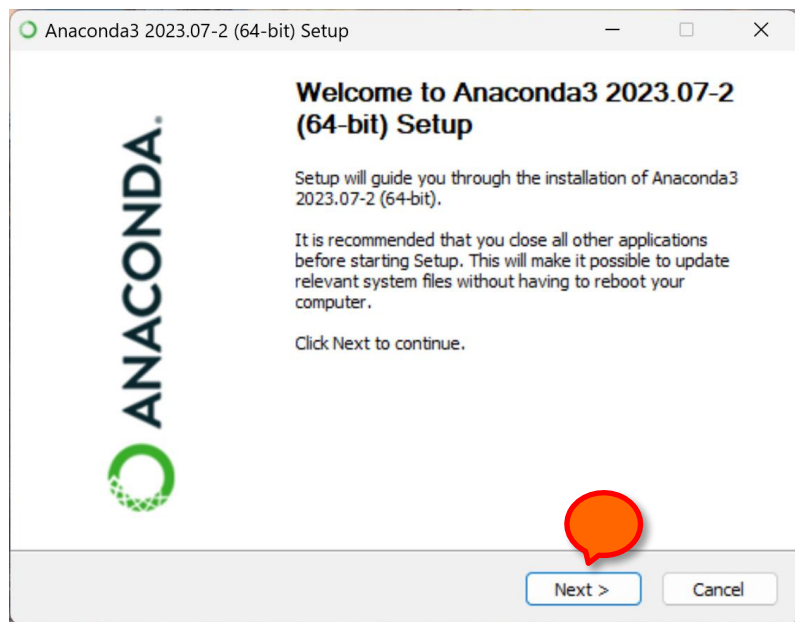


启动安装向导



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按图中提示操作





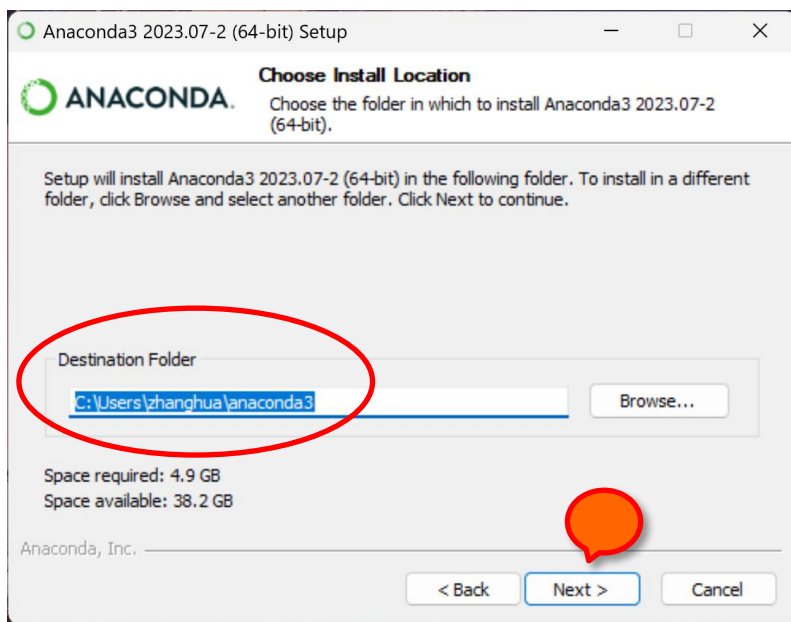
按照安装向导的步骤操作



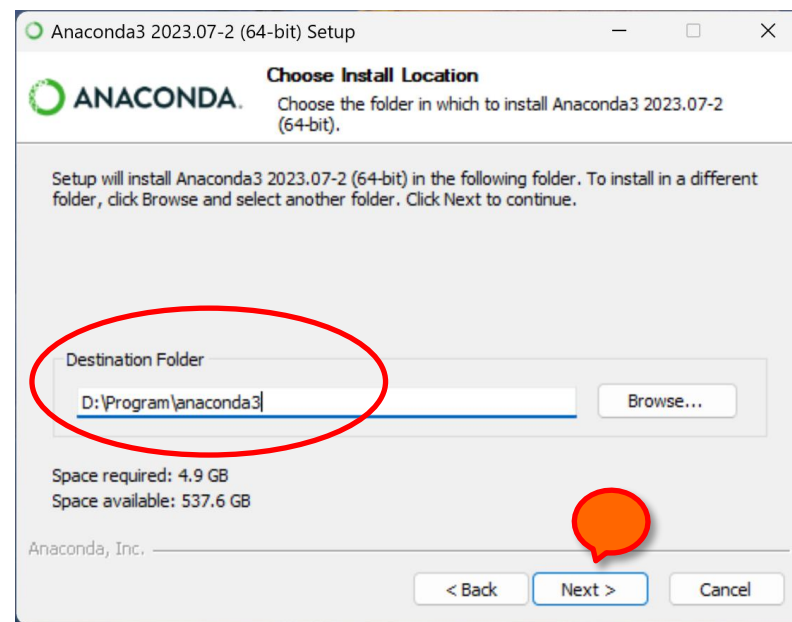
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选择安装位置

如果你的电脑上只有 C 盘，则使用默认安装位置，你不需修改安装位置



如果你的电脑上有 D 盘，则建议修改安装位置，图中位置仅供参考





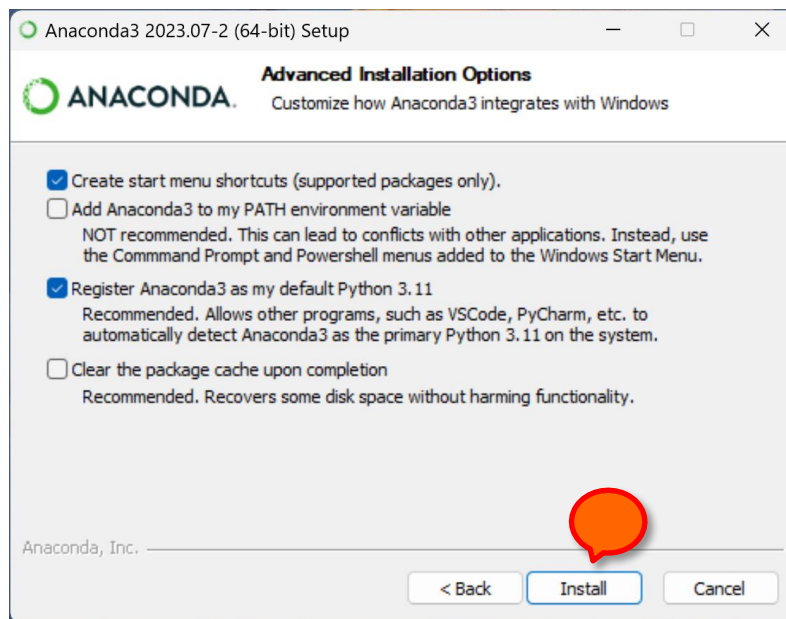
按照安装向导的步骤操作



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高级选项

建议直接使用默认选择，不做修改



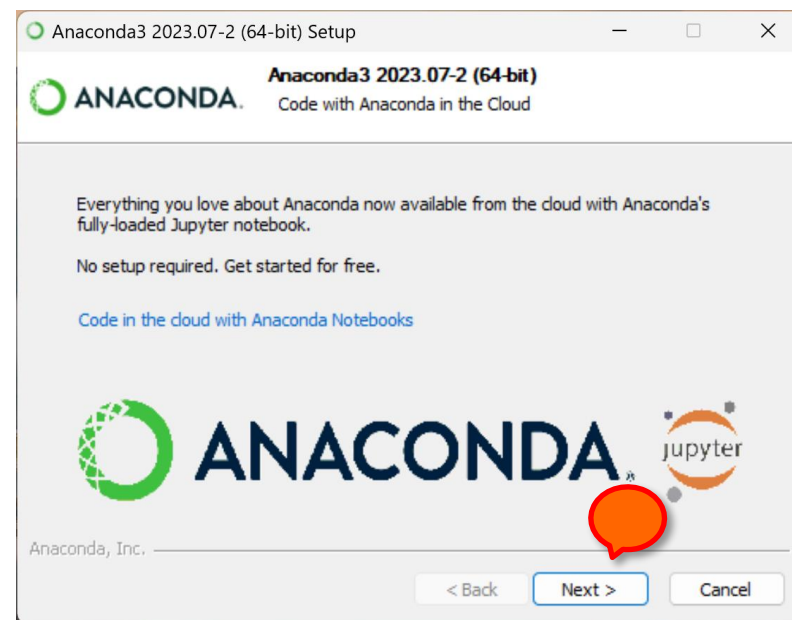
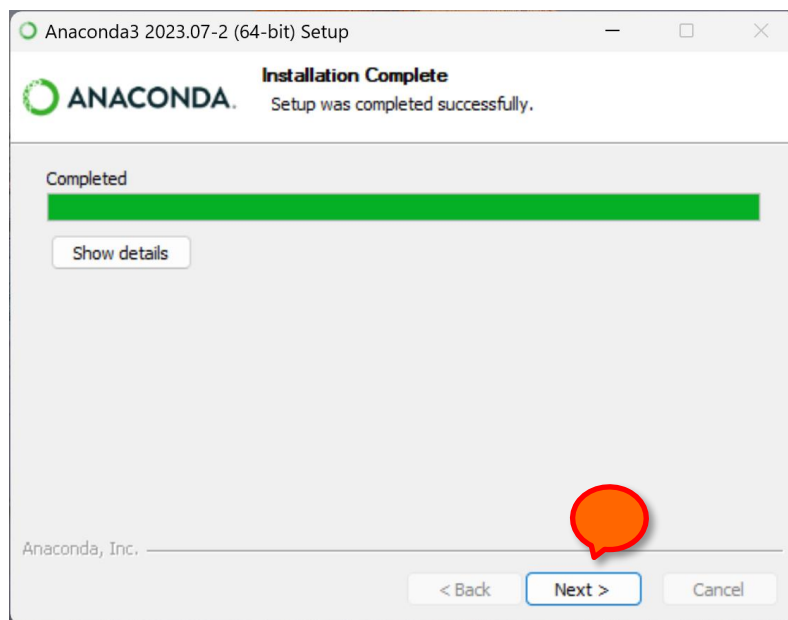
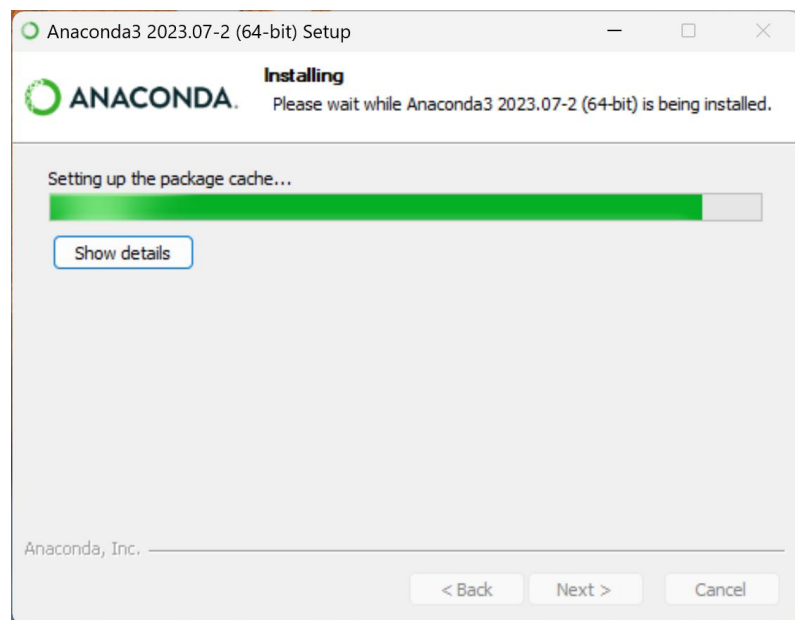


按照安装向导的步骤操作



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开始安装，等待完成...





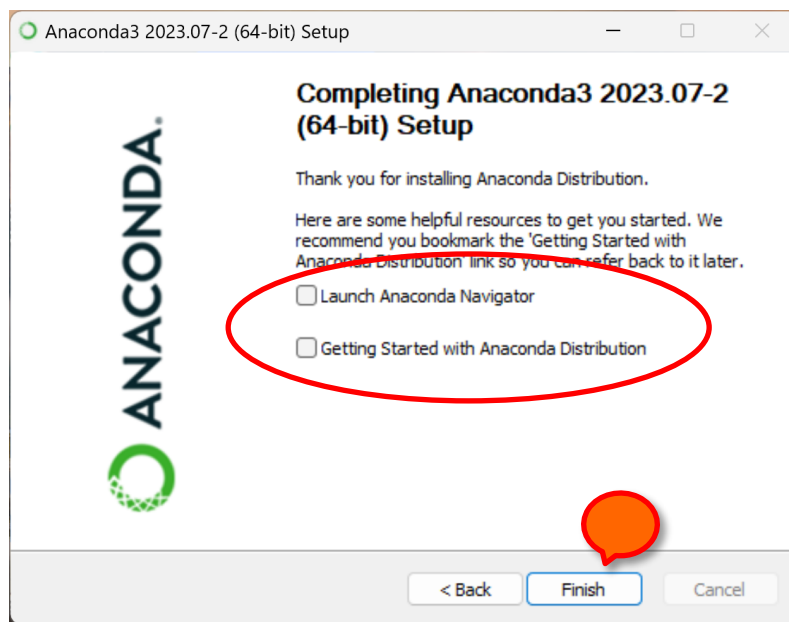
按照安装向导的步骤操作



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■ 安装完成

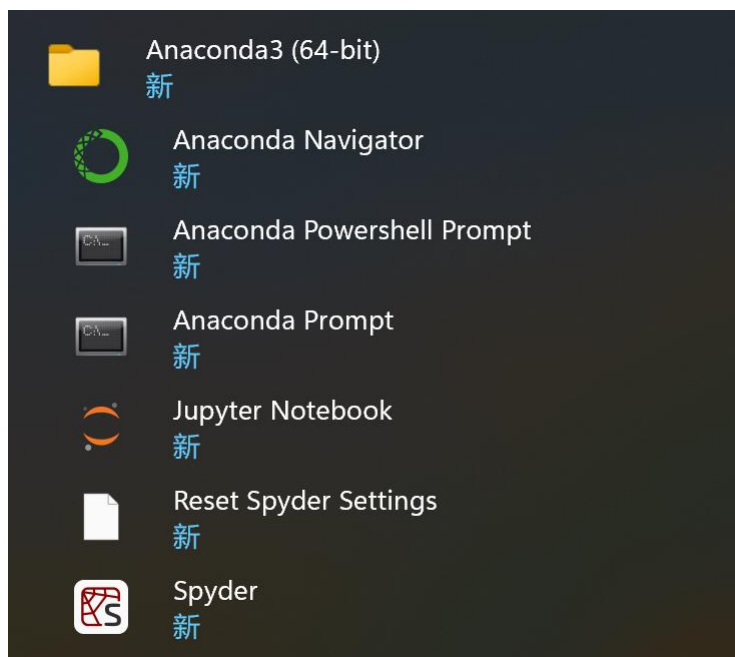
建议不要勾选图中两个选项





■ 检查程序组

✱ 在Windows 11开始菜单的“所有应用”中，找到下面的程序组及其菜单项





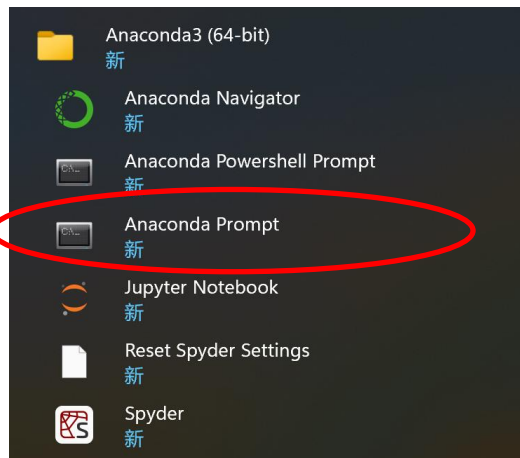
试用：Python解释器



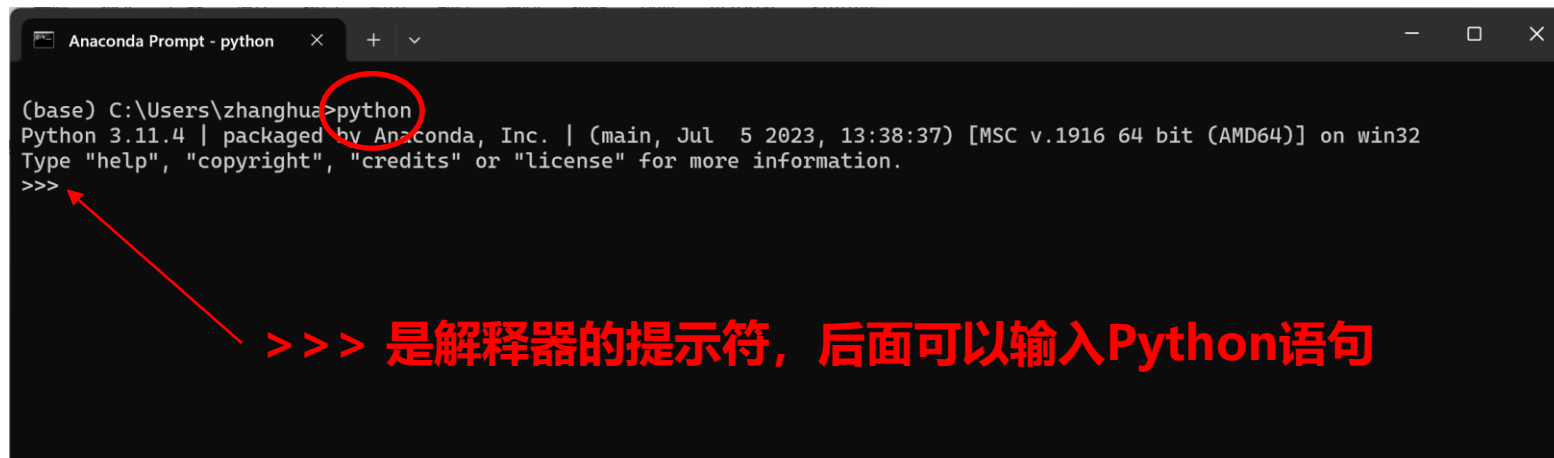
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启动Python解释器

点击 Anaconda Prompt



输入 python ，按键盘上的 Enter 键



>>> 是解释器的提示符，后面可以输入Python语句



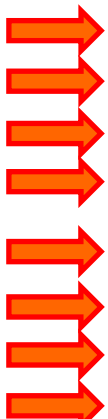
试用：Python解释器



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在Python解释器中输入并执行语句

在提示符后输入语句，然后按键盘上的 Enter 键执行语句



```
Anaconda Prompt - python
(base) C:\Users\zhanghua>python
Python 3.11.4 | packaged by Anaconda, Inc. | (main, Jul 5 2023, 13:38:37) [MSC v.1916 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("人生苦短，我学Python!")
人生苦短，我学Python!
>>> 123 + 456 - 789
-210
>>> 123 * 4 / 5
98.4
>>> 123 ** 99
795437446969572962215989474403487918968601125607596349321350197902962716256188267727046939206633805377147975587971037697
856066197182782519397377584138716051409888241865404815003470901166310031944013830701187
>>> 123 / 4
30.75
>>> 123 // 4
30
>>> (1 + 2 + 3) / 3
2.0
>>> import this
The Zen of Python, by Tim Peters

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Special cases aren't special enough to break the rules.
```




试用：Python解释器



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退出Python解释器

在提示符后输入语句 `exit()`，然后按键盘上的 Enter 键

```
Anaconda Prompt
30.75
>>> 123 // 4
30
>>> (1 + 2 + 3) / 3
2.0
>>> import this
The Zen of Python, by Tim Peters

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than *right* now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!
>>> exit()

(base) C:\Users\zhanghua>
```





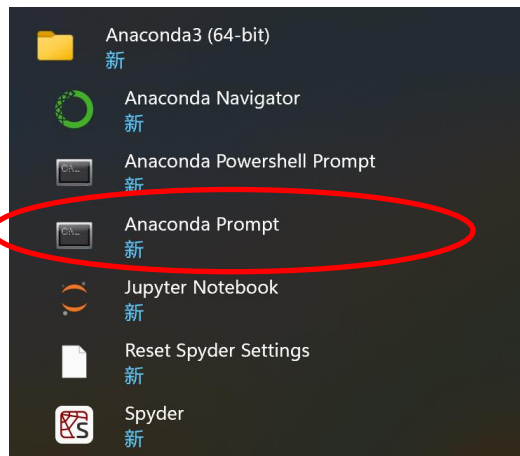
试用: IDLE



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启动IDLE

点击 Anaconda Prompt



输入 idle , 按键盘上的 Enter 键





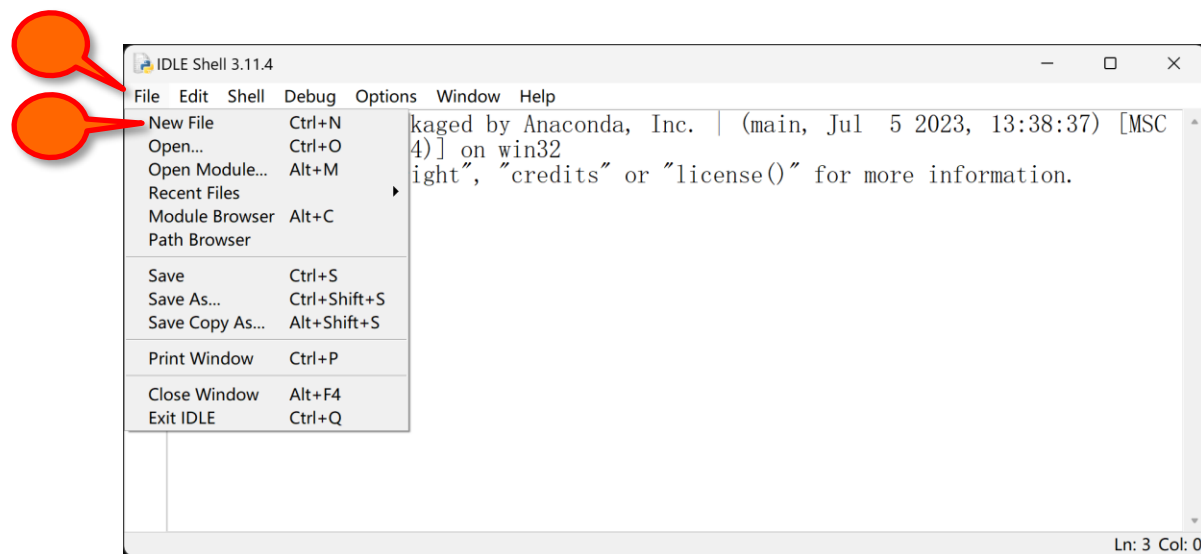
试用：IDLE



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创建程序文件，输入语句

打开菜单“File”，选择“New File”



在打开的编辑窗口中输入语句





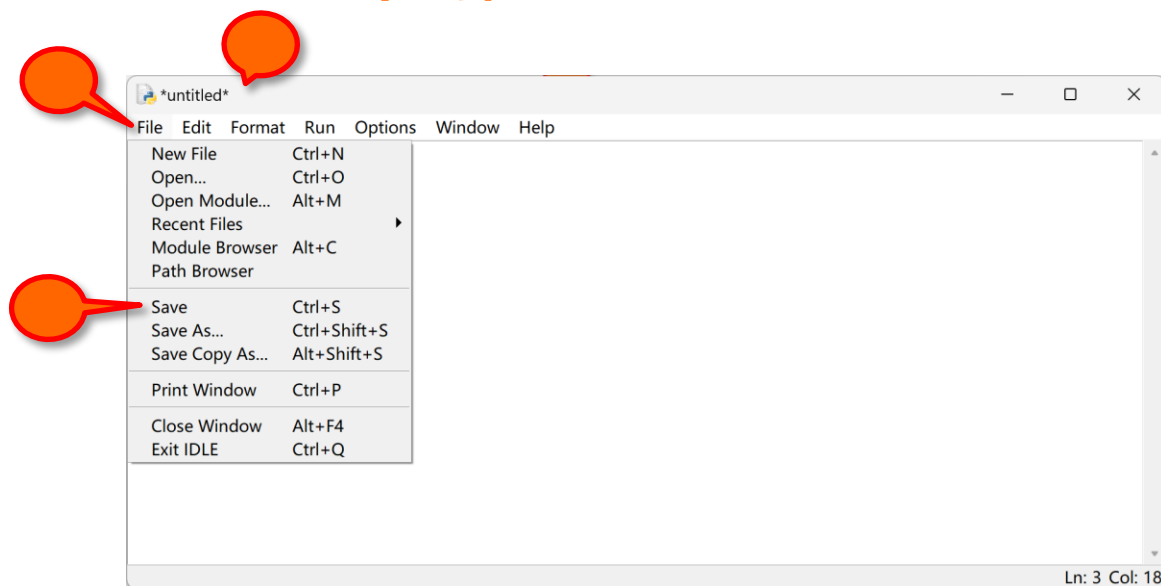
试用：IDLE



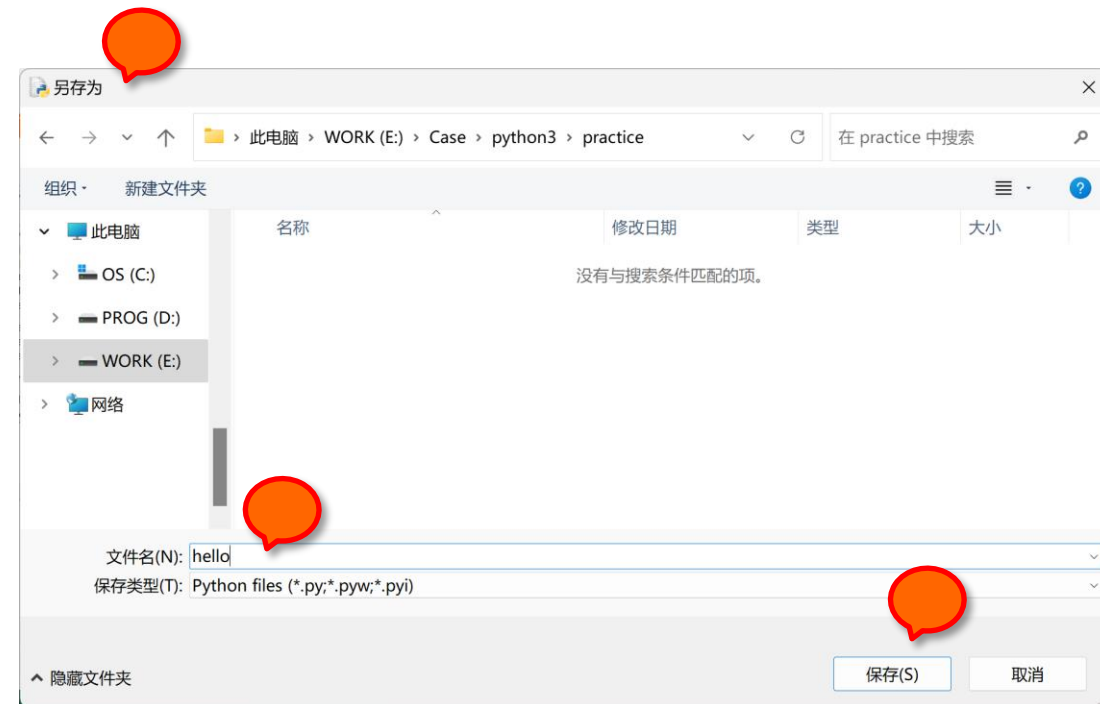
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保存程序文件

在“File”中选择“Save”



在打开的“另存为”对话框中选择位置，输入文件名，然后点击“保存”，创建 hello.py 文件





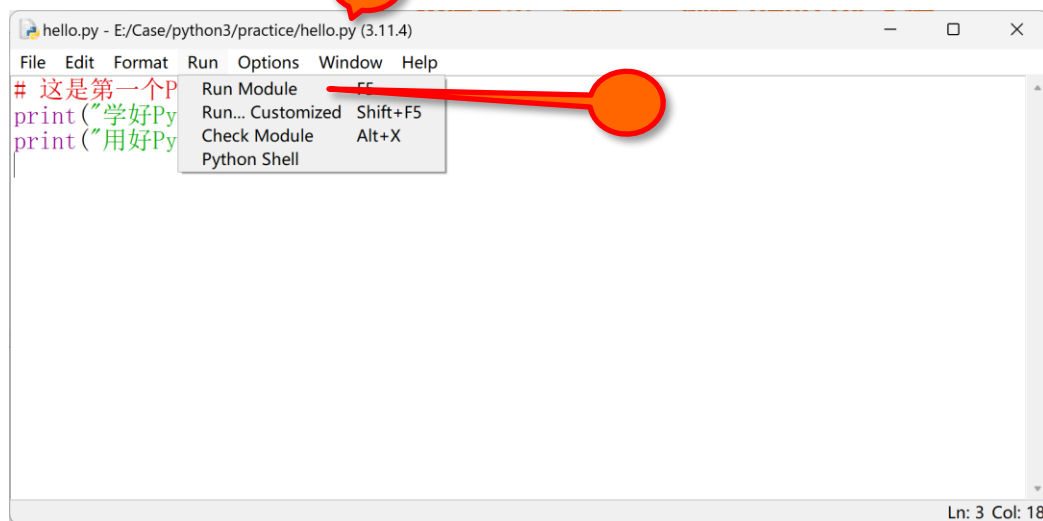
试用：IDLE



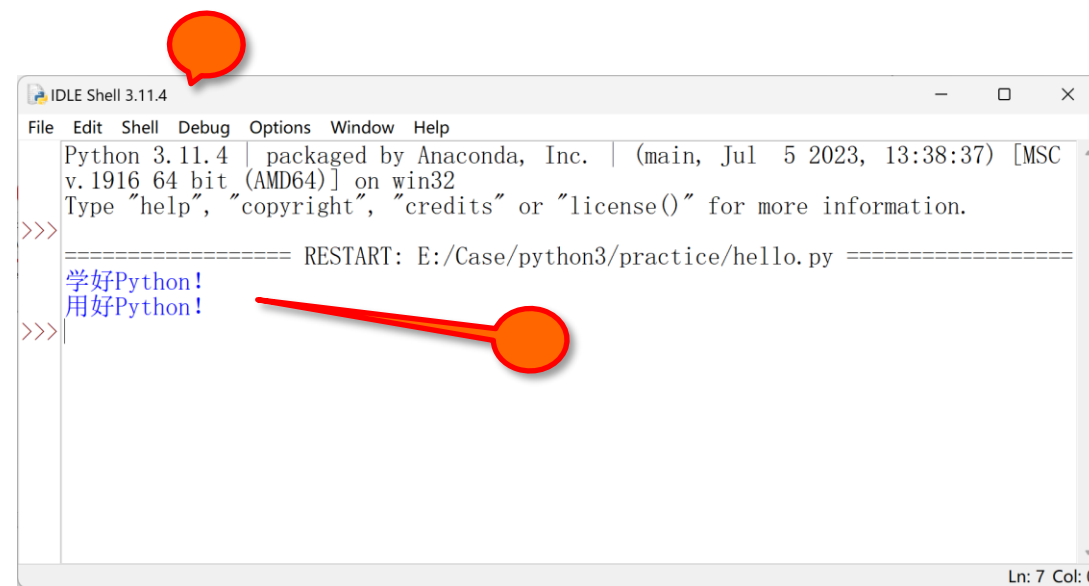
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■ 执行程序文件

在编辑窗口中，打开菜单“Run”，选择“Run Module”



在IDLE的解释器窗口中查看执行结果





试用: Jupyter Notebook

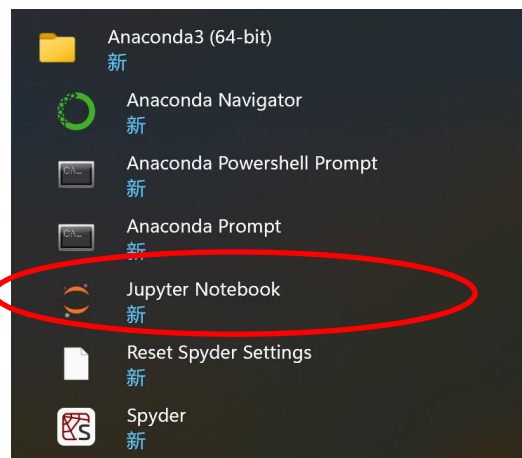


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启动Jupyter Notebook

老师将讲解具体使用方法

点击 Jupyter Notebook

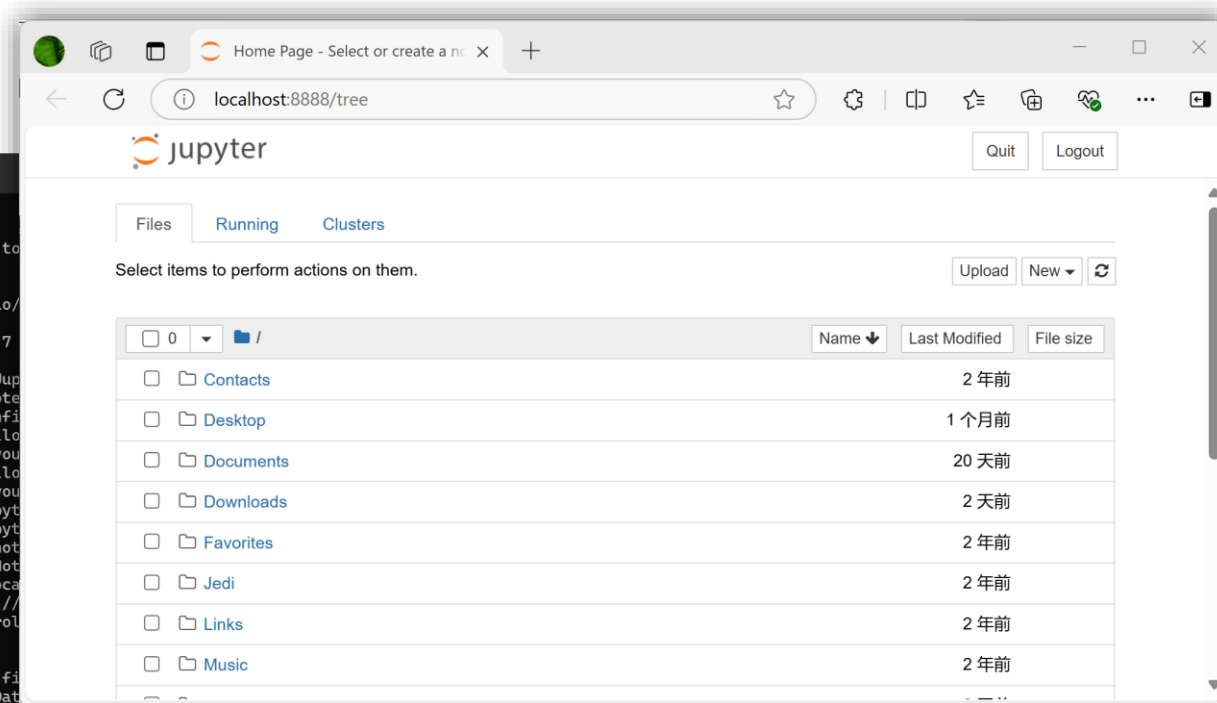


```
Jupyter Notebook

Read the migration plan to Notebook 7 to
https://jupyter-notebook.readthedocs.io/

Please note that updating to Notebook 7

[W 22:04:36.986 NotebookApp] Loading Jupyter Notebook
[W 2024-09-22 22:04:36.988 LabApp] 'notebook' is deprecated in
ServerApp. Be sure to update your configuration.
[W 2024-09-22 22:04:36.989 LabApp] 'allow_root_access' is deprecated in
ServerApp. Be sure to update your configuration.
[W 2024-09-22 22:04:36.989 LabApp] 'allow_root_access' is deprecated in
ServerApp. Be sure to update your configuration.
[I 2024-09-22 22:04:36.993 LabApp] Jupyter Notebook is running at
http://localhost:8888/ or http://127.0.0.1:8888/
[I 22:04:39.893 NotebookApp] Serving notebook
[I 22:04:39.893 NotebookApp] Jupyter Notebook is running at
http://localhost:8888/ or http://127.0.0.1:8888/
[I 22:04:39.893 NotebookApp] Use Control-C to stop the server and
quit the notebook.
[C 22:04:39.924 NotebookApp]
```

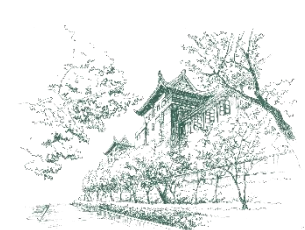




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02

在macOS中安装并试用



执行安装文件



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■ 在桌面或访达中找到并执行安装文件

- ✿ Anaconda3-2023.07-2-MacOSX-arm64.pkg

- ✿ 或者

- ✿ Anaconda3-2023.07-2-MacOSX-x86_64.pkg

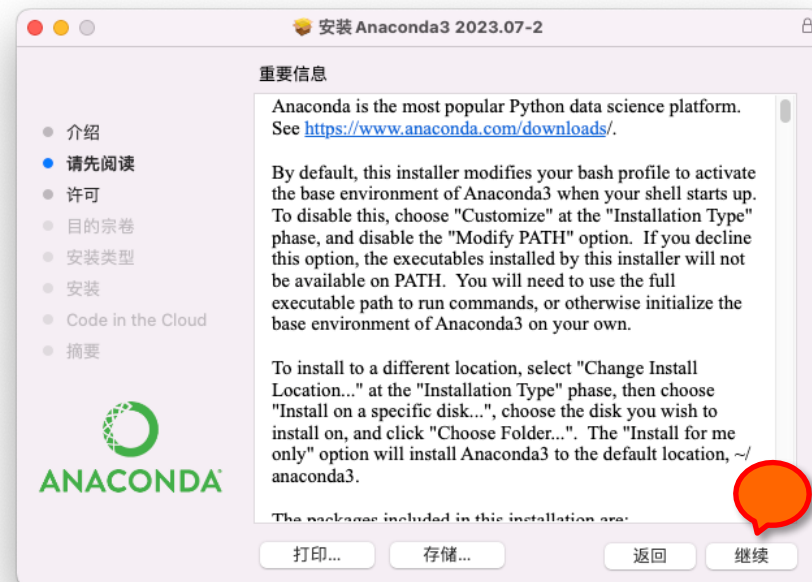


启动安装向导



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按图中提示操作



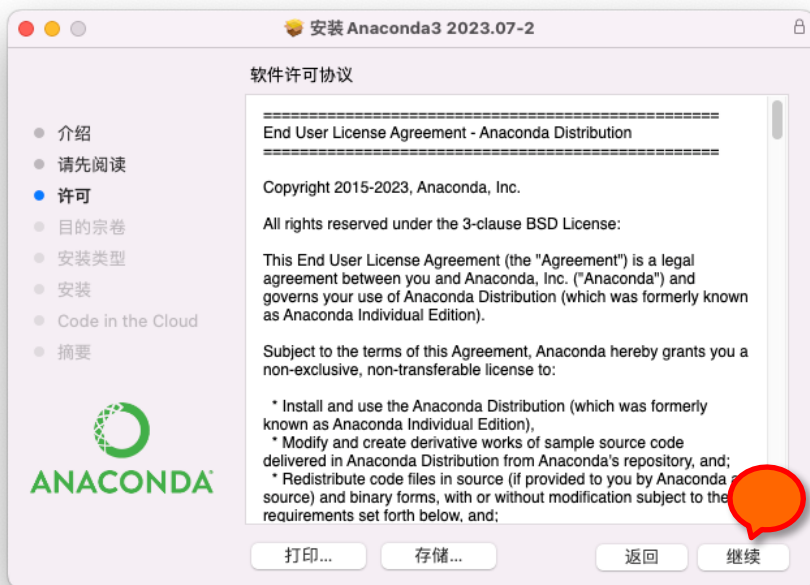


启动安装向导



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按图中提示操作





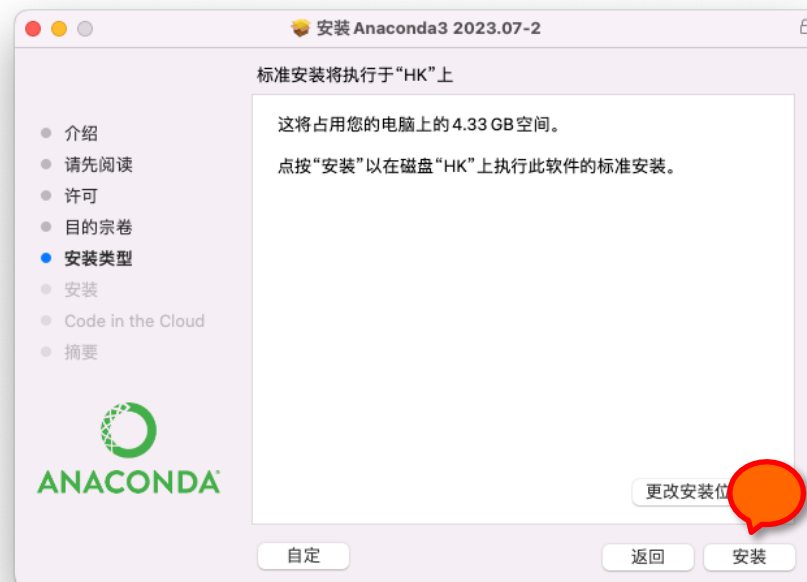
按照安装向导的步骤操作



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选择安装位置

建议使用默认安装位置，你不需更改安装位置





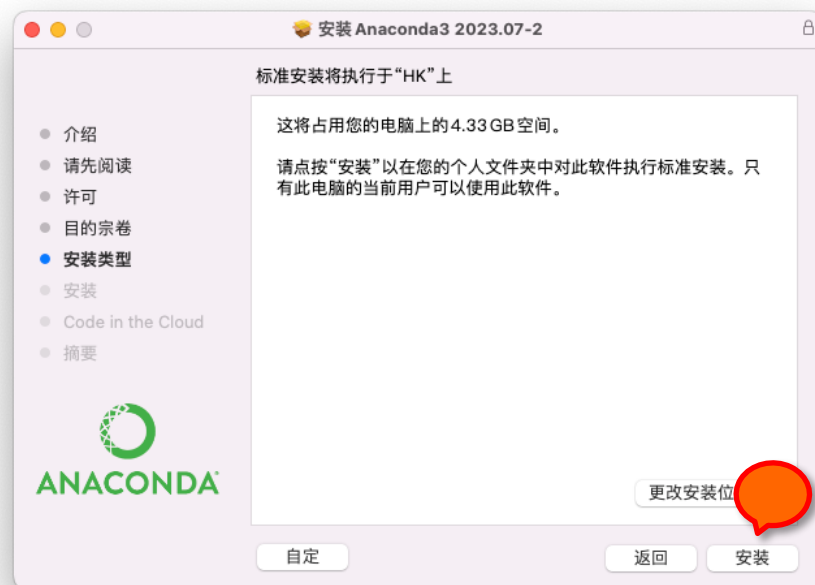
按照安装向导的步骤操作



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■ 安装类型

建议直接使用默认选择，不做修改



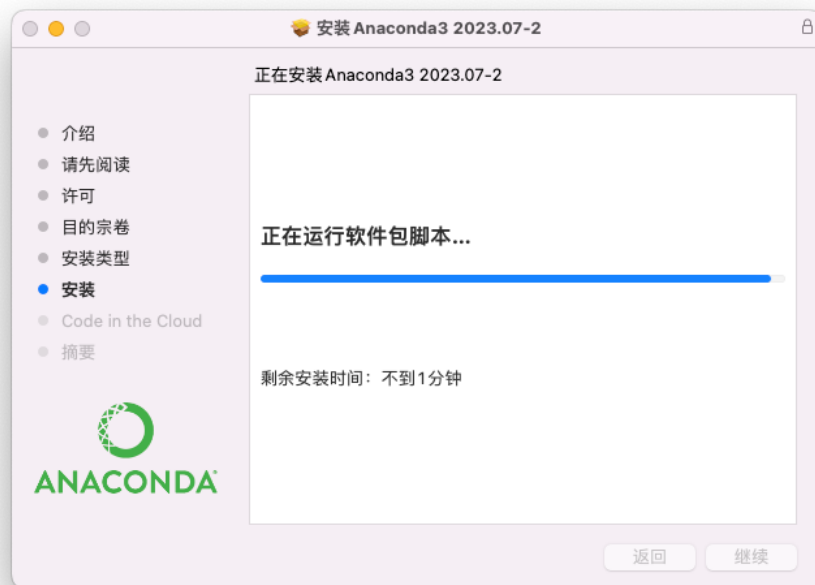


按照安装向导的步骤操作



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开始安装，等待完成...





按照安装向导的步骤操作



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■ 安装完成



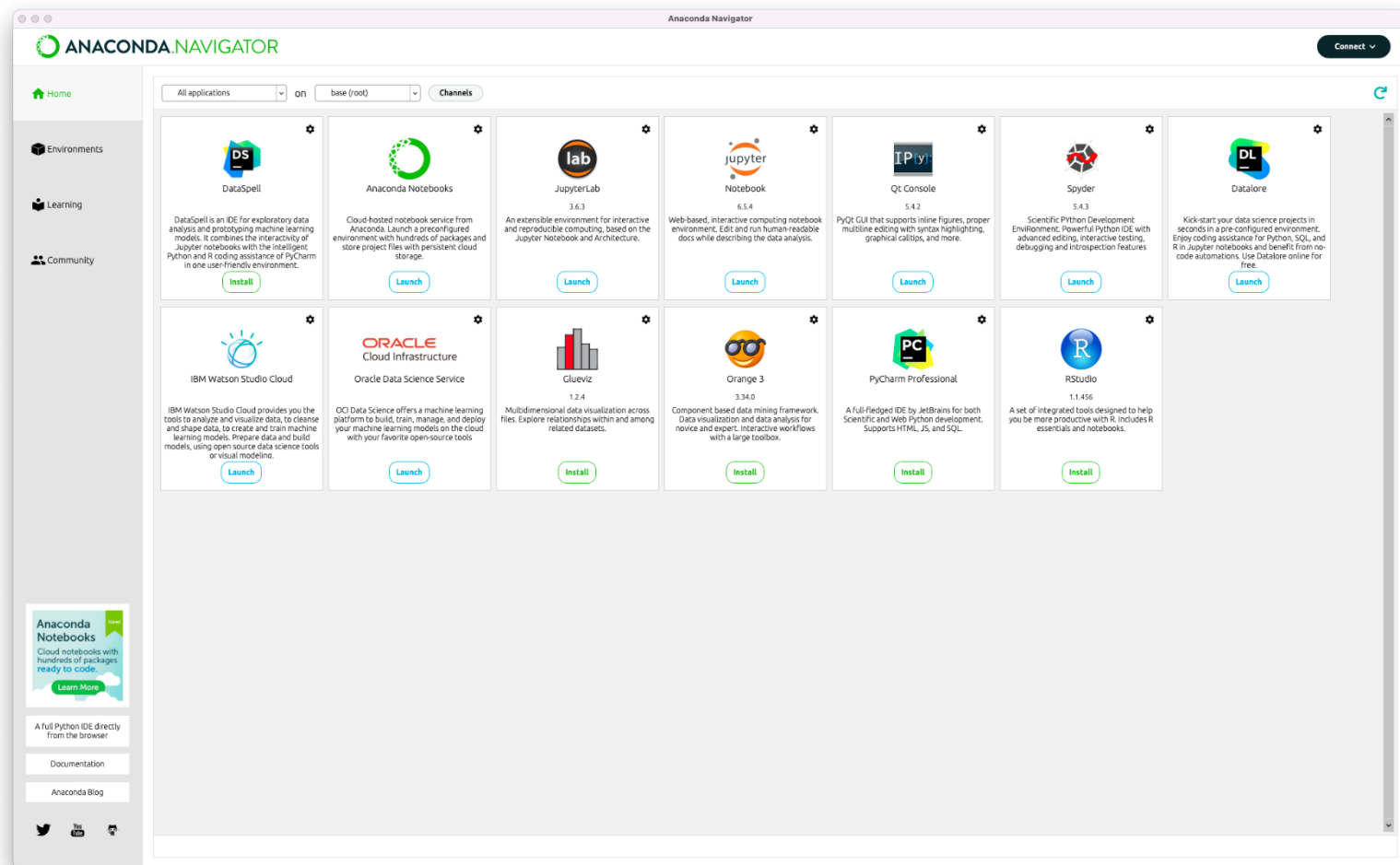


安装后的操作



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■ 安装成功后，会自动启动Navigator，建议直接关闭



如果出现下面的询问，选“好”





试用：Python解释器



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启动Python解释器

输入 `python`，按键盘上的 Enter 键。在较老的系统上，需要输入 `python3`

在启动台中找到并选择“
终端”（terminal）



```
zhanghua — python — 80x24
Last login: Thu Aug 24 13:53:32 on ttys001
[(base) zhanghua@ZhangHuadeMac-mini ~ % python
Python 3.11.4 (main, Jul 5 2023, 09:00:44) [Clang 14.0.6 ] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

>>> 是解释器的提示符，后面可以输入Python语句



试用：Python解释器



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在Python解释器中输入并执行语句

在提示符后输入语句，然后按键盘上的 Enter 键执行语句



```
zhanghua — python — 80x24
Last login: Thu Aug 24 13:53:32 on ttys001
[(base) zhanghua@ZhangHuadeMac-mini ~ % python
Python 3.11.4 (main, Jul 5 2023, 09:00:44) [Clang 14.0.6 ] on darwin
Type "help", "copyright", "credits" or "license" for more information.
[>>> print("人生苦短，我学Python! ")
人生苦短，我学Python!
>>> ]
```

可以参考Windows版中试用部分，输入并执行更多语句



试用：Python解释器



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■ 退出Python解释器

在提示符后输入语句 `exit()`，然后按键盘上的 Enter 键



试用：IDLE



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启动IDLE

启动terminal，输入 `idle3`，然后按键盘上的 Enter 键

```
zhanghua — zsh — 80x24
Last login: Thu Aug 24 14:23:08 on ttys000
(base) zhanghua@ZhangHuadeMac-mini ~ % idle3
```



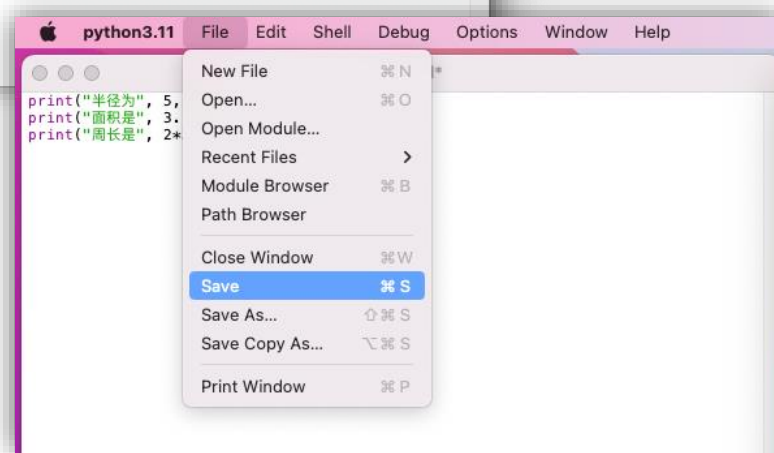
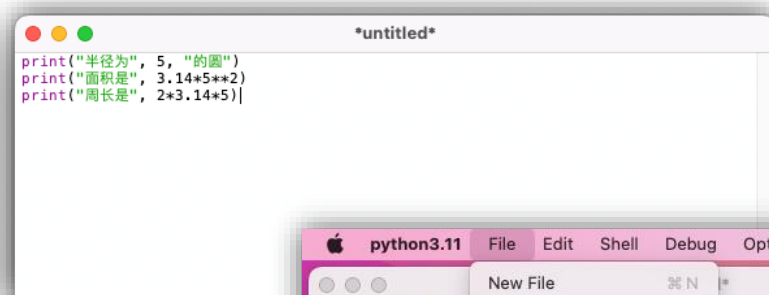
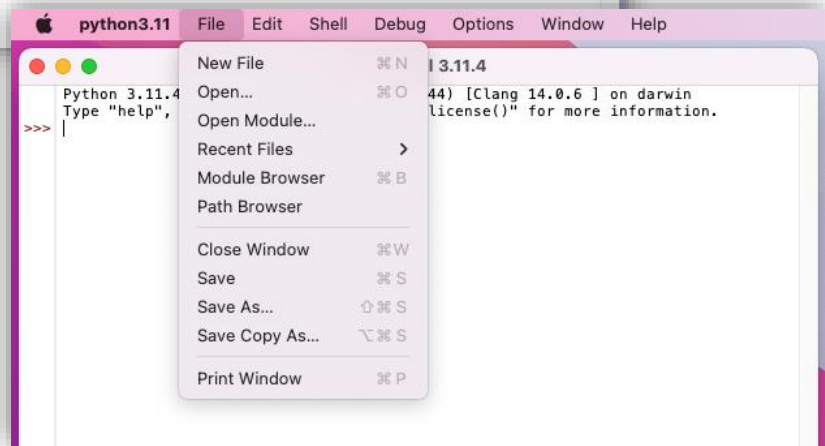
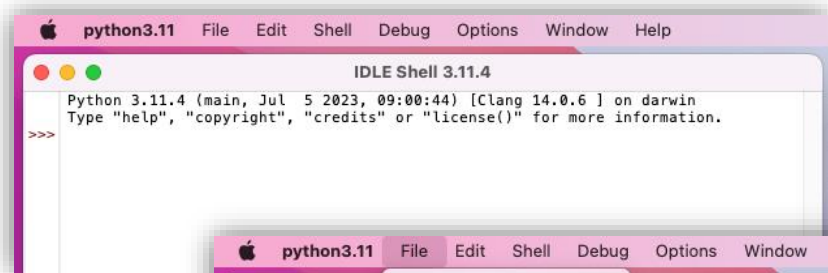

试用：IDLE



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■ 新建程序文件，输入语句，保存文件，执行程序

✳ 操作与Windows部分类似





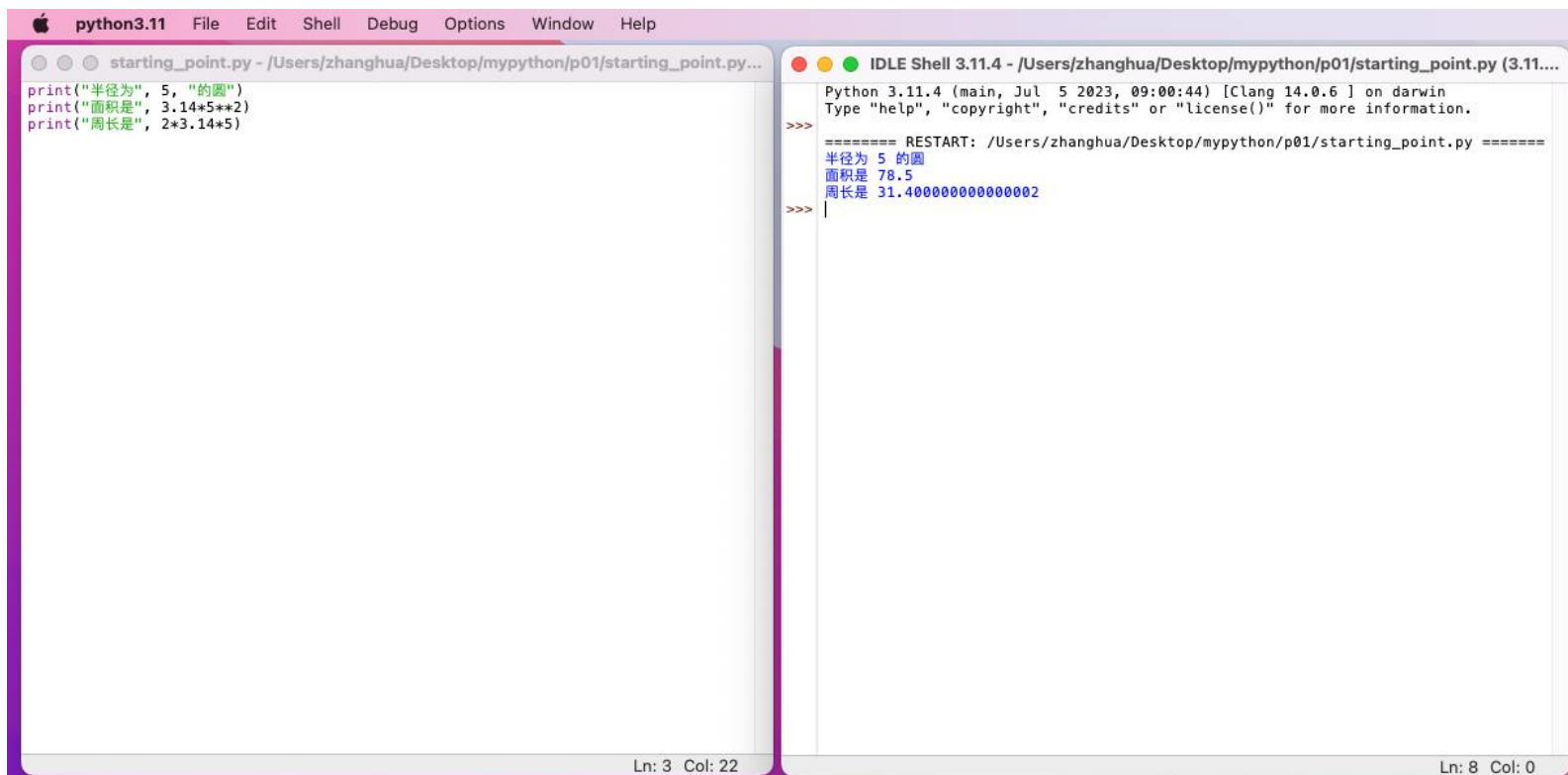
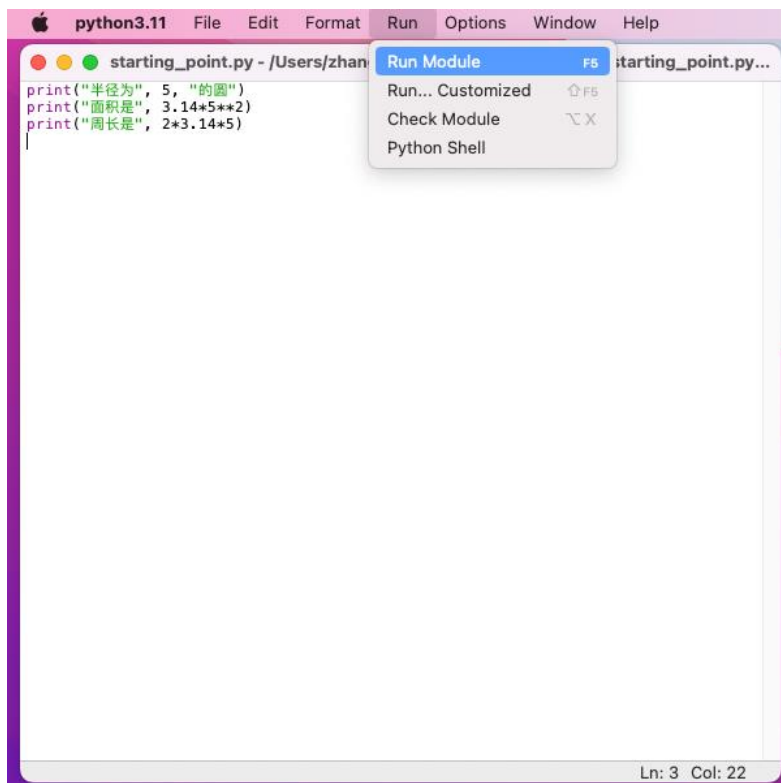
试用：IDLE



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■ 新建程序文件，输入语句，保存文件，执行程序

✳ 操作与Windows部分类似





试用：Jupyter Notebook



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启动Jupyter Notebook

启动terminal，输入 `jupyter notebook`，然后按键盘上的 Enter 键

```
zhanghua — zsh — 80x24
Last login: Thu Aug 24 14:04:13 on ttys000
(base) zhanghua@ZhangHuadeMac-mini ~ % jupyter notebook
```



试用：Jupyter Notebook



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启动Jupyter Notebook

老师将讲解具体使用方法

```
zhanghua — jupyter-notebook — 138x38
Last login: Thu Aug 24 14:04:13 on ttys000
(base) zhanghua@ZhangHuadeMac-mini ~ % jupyter notebook

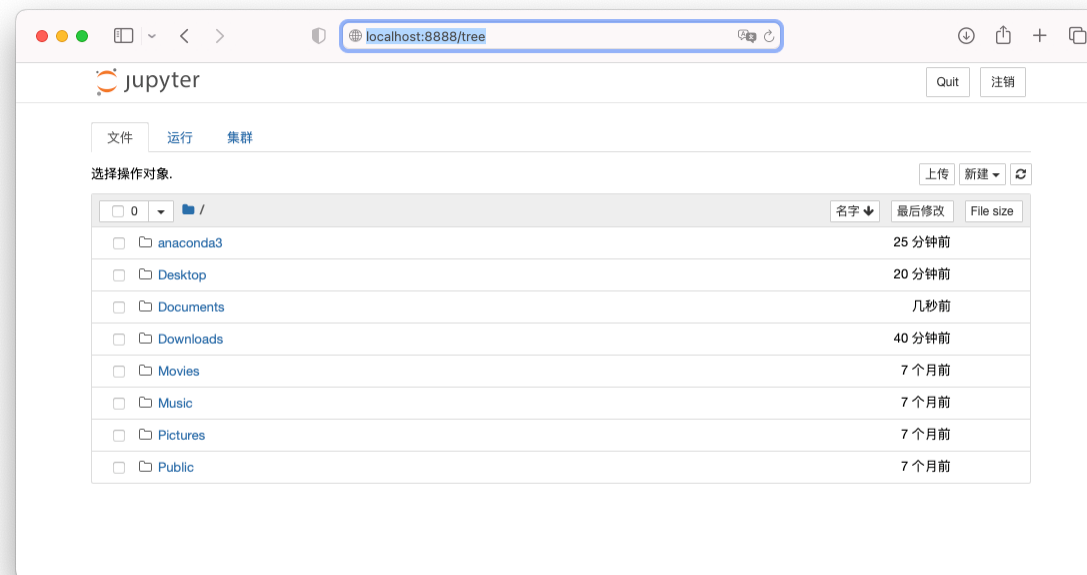
[Out]
[In]

Read the migration plan to Notebook 7 to learn about the new features and the actions to take if you are using extensions.
https://jupyter-notebook.readthedocs.io/en/latest/migrate_to_notebook7.html

Please note that updating to Notebook 7 might break some of your extensions.

[W 14:10:54.670 NotebookApp] Loading JupyterLab as a classic notebook (v6) extension.
[I 2023-08-24 14:10:54.677 LabApp] JupyterLab extension loaded from /Users/zhanghua/anaconda3/lib/python3.11/site-packages/jupyterlab
[I 2023-08-24 14:10:54.677 LabApp] JupyterLab application directory is /Users/zhanghua/anaconda3/share/jupyter/lab
[I 14:10:56.735 NotebookApp] notebooks 运行所在的本地路径: /Users/zhanghua
[I 14:10:56.735 NotebookApp] Jupyter Notebook 6.5.4 is running at:
[I 14:10:56.735 NotebookApp] http://localhost:8888/?token=d9c3a5f16d41052c68ab2687a296a6747088a3761549af46
[I 14:10:56.735 NotebookApp] or http://127.0.0.1:8888/?token=d9c3a5f16d41052c68ab2687a296a6747088a3761549af46
[I 14:10:56.736 NotebookApp] 使用 Control-C 停止此服务器并关闭所有内核（连续操作两次便可跳过确认界面）。
[C 14:10:56.739 NotebookApp]

To access the notebook, open this file in a browser:
file:///Users/zhanghua/Library/Jupyter/runtime/nbsrvr-5704-open.html
Or copy and paste one of these URLs:
http://localhost:8888/?token=d9c3a5f16d41052c68ab2687a296a6747088a3761549af46
or http://127.0.0.1:8888/?token=d9c3a5f16d41052c68ab2687a296a6747088a3761549af46
0.00s - Debugger warning: It seems that frozen modules are being used, which may
0.00s - make the debugger miss breakpoints. Please pass -Xfrozen_modules=off
0.00s - to python to disable frozen modules.
0.00s - Note: Debugging will proceed. Set PYDEVD_DISABLE_FILE_VALIDATION=1 to disable this validation.
```





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谢谢

THANK YOU

